IFEANYI DESMOND DIKE

Senior Software Engineer | Voice AI & Distributed Systems

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PROFILE

Software engineer with **9+ years of experience** building backend services, real-time media and AI products. Owns voice infrastructure from telephony and streaming speech through LLM orchestration, deployment and evaluation. Recent work includes on-demand call capacity, distributed scheduling and multilingual voice agents. Strong Python, Go and TypeScript background, with production systems supporting up to 10,000 concurrent connections.

TECHNICAL SKILLS

Languages & backend: Python, asyncio, Go, TypeScript, FastAPI, gRPC, REST, PostgreSQL, Redis, MongoDB, RQ background workers.

Voice & AI: Pipecat, Twilio, WebSockets, WebRTC, streaming STT/TTS, VAD, turn-taking, LLM tool use, RAG, evaluation, PyTorch, Hugging Face.

Infrastructure: Fly.io, Docker, Kubernetes, AWS, GCP, Azure, CI/CD, observability, distributed systems and multi-tenant services.

PROFESSIONAL EXPERIENCE

DataStrut AI | Founding AI Engineer / Senior Software Engineer

Aug 2025 - Present | Remote

- Own the production voice platform across telephony, streaming speech, LLM orchestration and post-call processing; migrated managed voice infrastructure to a self-hosted Python/Pipecat stack.
- Scaled from fixed shared capacity to an on-demand Fly Machines pool with per-call isolation, atomic Redis reservations, capacity-aware routing and warm capacity reserved for inbound calls.
- Built safe machine drain, resume and release lifecycles that account for call setup and post-call work; fenced delayed callbacks to protect newer reservations.
- Separated outbound scheduling from report generation with dedicated RQ workers. Added asynchronous, idempotent call triggers and capacity-aware retries to avoid duplicate dials and queue contention.
- Designed production-informed load tests and CPU, memory, Redis-connection and report-throughput models to identify bottlenecks and guide capacity planning.
- Delivered English/Spanish voice workflows across speech recognition, synthesis, prompts, recovery messages and post-call reporting.
- Built response-admission controls, echo-aware turn-taking and deterministic regression tests; added playback-aligned traces and recordings to distinguish generated speech from what callers heard.

Cenphi | Co-Founder and Lead Developer

Mar 2025 - Present | Remote

- Built an asynchronous video-production pipeline with Python, FastAPI, Whisper, diarization, LLMs and FFmpeg; added job orchestration, provider fallback and live progress updates.
- Developed segment-based story composition and artifact tracking, reducing manual review by approximately 80%.

PROFESSIONAL EXPERIENCE - CONTINUED

Fynance Hub | Technical Lead

Oct 2024 - Present | Remote, part-time

- Lead backend architecture and delivery using Go, Python, gRPC, PostgreSQL and Redis for real-time financial services.
- Built retrieval-augmented recommendations and automated voice-verification workflows, from service design through deployment.

Arbor Platform Inc. | Senior Software Engineer

Feb 2023 - Oct 2024 | Remote

- Built Claude/OpenAI-powered video workflows that reduced production time by 60% and improved throughput by 80%.
- Delivered text-to-video, transcription, content-studio and asset-management features across React/TypeScript interfaces, backend APIs and cloud services.

Howdy Learning | Senior Software Development Engineer

May 2020 - Mar 2023 | Remote

- Reduced WebSocket network traffic by more than 99% through real-time communication optimizations.
- Built live-video infrastructure supporting up to 10,000 concurrent connections and chat serving approximately 100 communities.
- Delivered embedded video, voice, chat and single sign-on integrations for more than 100 customers.

Bujuwa Nigeria Limited | Software Developer

Jul 2018 - Dec 2019 | Abuja, Nigeria

- Delivered web and mobile assessment systems, covering requirements, application architecture and implementation.

OPEN SOURCE & SELECTED ENGINEERING WORK

Pipecat: Diagnosed and reported stale-generation failures in the Anthropic integration, with reproduction evidence in #5035 and #5036; maintainers resolved #5036 through upstream PR #5041.

Acoustic Turn-over Rate (ATR): Developed an ASR-independent voice-agent evaluation metric. On controlled audio, ASR-based estimates diverged by up to 23.5 percentage points; three VAD backends agreed within 7.6 points. Code | Research artifact.

Full-Duplex-Bench reproduction: Investigated ASR sensitivity in voice-agent evaluation, finding up to 62x metric variation across ASR backends on the same audio. Code and results.

TubeForge: Built an open-source video workflow backend with Python, FastAPI, SQLAlchemy, FFmpeg, WebSocket progress and provider fallback. Source code.

Technical writing: Publish practical investigations of voice-agent concurrency, interrupted speech and turn-taking at ifeanyidike.com/blog.

EDUCATION

M.Sc., Financial Engineering | WorldQuant University | 2026 - Present (in progress)

B.Tech., Applied Mathematics, First Class | Federal University of Technology, Akure | 2018
GPA: 4.63/5.00. National university mathematics competition: Gold (2017), Bronze (2016).